

Chacha Chen

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I am interested in the broad area of human-centered machine learning, at the intersection of machine learning and human-computer interaction. I am passionate to work on topics but not limited to: interpretable ML, human-AI decision making, and human-AI cooperation.

EDUCATION **University of Chicago** 2021 – present
 Department of Computer Science
 Ph.D. in Computer Science
 Advisor: [Chenhao Tan](#)

Pennsylvania State University 2019 – 2020
 College of Information Sciences and Technologies
 Master in Informatics
 Advisor: [Ting-Hao \(Kenneth\) Huang](#), [Zhenhui \(Jessie\) Li](#)

Shanghai Jiao Tong University 2015 – 2019
 Department of Computer Science
 Bachelor of Science in Computer Science

PROFESSIONAL **University of Chicago**, Research Assistant 2021 – present
EXPERIENCE *with [Chenhao Tan](#)*
 A theoretical framework of human understanding and machine explanations [\[P.1\]](#).
 Human-AI decision making survey [\[P.2\]](#).

IQVIA, Research Intern 2020
 with [Cao \(Danica\) Xiao \(IQVIA\)](#), [Fenglong Ma \(PSU\)](#) and [Jimeng Sun \(UIUC\)](#).
 Uncertainty based health risk prediction with deep probabilistic models [\[P.5\]](#).

Pennsylvania State University, Research Assistant 2019 – 2020
 with [Kenneth Huang](#)
 Visual story telling [\[P.3\]](#)
 Slot filling in noisy task [\[W.4\]](#)
 HCI related projects [\[W.1\]](#)[\[W.2\]](#)

Pennsylvania State University, Research Assistant 2019 – 2020
 with [Zhenhui \(Jessie\) Li](#)
 Reinforcement learning for traffic signal control [\[P.8\]](#)[\[P.9\]](#)[\[P.10\]](#).
 Learning to simulate [\[P.4\]](#)[\[P.6\]](#)[\[P.7\]](#)
 Physics-informed GAN[\[W.3\]](#)

PUBLICATIONS  [Google scholar](#)

Manuscripts and Pre-prints

P.1 **Chacha Chen***, Shi Feng*, Amit Sharma, Chenhao Tan.
Human Understanding and Machine Explanations
Submitted to FAccT 2022, under review.

- P.2 Vivian Lai, **Chacha Chen**, Alison Smith-Renner, Vera Liao, and Chenhao Tan.
Towards a Science of Human-AI Decision Making: A Survey of Empirical Studies
In arXiv:2112.1147.
- P.3 Chi-Yang Hsu, Yun-Wei Chu, Vincent Chen, Kuan-Chieh Lo, **Chacha Chen**, Ting-Hao Huang, Lun-Wei Ku.
Learning to Rank Visual Stories From Human Ranking Data
Submitted to ACL November 2021, under review.

Conference Papers

- P.4 Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Tao Wen, Zhenhui Li. Knowledge-based Residual Learning. In Proceedings of the The 30th International Joint Conference on Artificial Intelligence (**IJCAI 2021**), Aug 2021.
- P.5 **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao, UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data, In Proceedings of The Web Conference 2021 (**WWW 2021**), Ljubljana, Slovenia, April 2021.
- P.6 Guanjie Zheng, Chang Liu, Hua Wei, **Chacha Chen**, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**), Chania, Greece, April 2021.
- P.7 Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**), Ghent, Belgium, Sep 2020. *Best Applied Data Science Paper Award*
- P.8 **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**), New York, United States, February 2020. (Acceptance rate: $1591/7737=20.6\%$)
- P.9 Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**), Beijing, China, November 2019. (Acceptance rate: $200/1030=19.4\%$)
- P.10 Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**), Anchorage, AK, USA, Aug 2019. (Acceptance rate: $170/1200=14.2\%$)

Workshop Papers

- W.1 Tongan Cai*, **Chacha Chen***, Ting-Hao K. Huang, Frank E. Ritter, What Makes A Good Reference Manager? A Quantitative Analysis of Bibliography Management Applications, Asian CHI Symposium 2021 (**AsianCHI2021**). *Best Paper Award*
- W.2 Jiaqi Wang, Hua Shen, **Chacha Chen**, Frank E. Ritter, Are Learners Satisfied with their MOOC Experiences? Assessing and Improving Online Learners' Interactions, Asian CHI Symposium 2021 (**AsianCHI2021**).
- W.3 **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning (**IIBPSL, NeurIPS 2020**).

W.4 **Chacha Chen**, Chieh-Yang Huang, Yaqi Hou, Yang Shi, Enyan Dai, Jiaqi Wang, Joint Event Multi-task Learning for Slot Filling in Noisy Text. In Proceedings of the Workshop on Noisy User-generated Text (**W-NUT, EMNLP 2020**).

COMMUNITY ACTIVITIES & SERVICE	Conference Program Committee Member/Reviewer • AAAI 2021: The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21) • W-NUT: The 6th Workshop on Noisy User-generated Text with EMNLP 2020 • NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning
TEACHING EXPERIENCE	Pennsylvania State University Teaching Assistant Fall 2020 IST 261 (Java Course): Application Development Design Studio I Prof. Kenneth Huang
AWARDS	AAAI-20 VISA Research Scholarship Feb. 2020 AAAI-20 Student Travel Grant Feb. 2020 Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i> Aug. 2020 Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i> Jun. 2019 Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i> Sep. 2016